MIL-STD-652D (AR)
NOTICE 3
27 October 1982

MILITARY STANDARD

PROPELLANTS, SOLID FOR CANNONS

REQUIREMENTS AND PACKING

To all Holders of MIL-STD-652D (AR)

1. The following pages of MIL-STD-652D (AR) have been revised and supersede the pages listed:

New Page	Date	Superseded Page	Date
6	27 October 1982	6	13 Mar 81
7	27 October 1982	7	5 Oct 79
8	27 October 1982	8	5 Oct 79

2. Retain this notice and insert before Table of Contents.

3. Holders of MIL-STD-652D (AR) will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1376-A205)

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4.3.1.1 Immediately prior to filling all containers listed in 4.3.1 shall withstand a 1/2 to 1 psi air pressure test for a minimum of 15 seconds without leakage. The containers shall be tested by a method satisfactory to the contracting officer's representative.

4.3.1.2 When replacing cover gaskets for the M2 Steel Box, Dwg. 76-4-55, solid rubber gaskets as described on Dwg. 138441 for the MK7 Packing Box (Navy) may be used in lieu of Gasket Part No. 76-4-55H. Solid rubber gaskets shall comply with Specification MIL-R-3065 and RN-715 or RS-715 of Standard MIL-STD-417.

4.3.1.3 Calibration. The amount of propellant selected for use as Master Calibration Lot or Reference Calibration Lot in accordance with TECOM Regulation 702-1 shall be packed in Level A containers (see 4.3.1).

4.3.1.4 Level A packing. The propellant M2, M5, M9, M10, M26 and M26E1 shall be packed in Level A containers, unless otherwise specified by the Contracting Office (see 4.3.1).

4.3.2 Level B. The propellant shall be packed as specified in 4.3.1.

4.3.3 Level C. The propellant shall be packed in fiber drums as described in 4.3.3.1 only when intended for assembly into an end item or subassembly within twelve months of the shipping date. Fiber drums are approved for truck or trailer on flat car (TOFC) shipment only and for storage not exceeding two years.

4.3.3.1 Fiber drums. Fiber drums shall comply with DOT Specification 21C, 250 pounds, MINIMUM, Code of Federal Regulations, Title 49, parts 100-199, and the following additional requirements. Size shall be 15 1/2 $\pm$ 1/2 inches in diameter by 26 $\pm$ 1 inches in height, inside dimensions. The drum shall have a 24 gauge steel cover with rubber gasket, lever locking band with provision for sealing wire and wide bottom chime (2 inch minimum formed height). All metal parts shall be hot-dipped galvanized. Top and bottom chime shall be 24 gauge steel and shall be welded. The body shall be wound with a hot melt or thermoplastic adhesive. The bottom shall be a waterproof laminated fiberboard. Body and bottom disc shall also have a laminated aluminum foil barrier. The bottom crimp shall be caulked. The finished drum with closure assembled shall be moistureproof and leak tight. The fiber drums may be reused if the drums comply with the inspection requirements of 4.4.1.3.

4.3.3.2 Alternative fiber drum. Alternatively, fiber drums shall be constructed as specified in 4.3.3.1 except that a layer of aluminum foil 0.010 thick shall be laminated to the inside of the body and the aluminum foil between the layers of Kraft paper in the body shall not be required.

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4.3.3.3 Marking. Drums shall be marked on the sidewall only with the same information as required for the side of the box by Dwg. 8858848. Alternatively, marking may be placed on a commercial water resistant label, securely and completely adhered to the side wall. The label stock shall be white, tan or kraft color. All marking shall be with black ink using letters approximately one half inch high.

4.3.4 Palletization. Level A shipments shall be palletized when specified by the procuring activity. Palletization is not required for Level B shipments.

4.4 Sampling for testing

4.4.1 Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105. Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

4.4.1.1 Container prior to filling (as applicable) (see Drawings 76-4-46, 76-4-53, 76-4-56, 138439, 7549033, and 9256486.

Categories		Defects	Method of Inspection
Critical:	None defined		
Major:	Class AQL 1.00 percent		
101		Foreign material, propellant or corrosion	Visual
102		Gasket missing or damaged	Visual
103		Holes in cover or end	Visual
104		Locking device damaged	Visual
105		Bare areas on exterior coating of metal container, the sum of which is in excess of 1/2 square inch	Visual

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Minor: Class AQL 6.50 percent

201	Protective finish incomplete	Visual
202	Wood split terminating at edge of board	Visual
203	Board broken or piece missing	Visual
204	Loose boards	Visual
205	Nails or staples protruding or loose	Visual
206	Split boards insufficiently nailed	Visual
207	Wood rot	Visual
208	Large dents or damaged seam	Visual

4.4.1.2 Fiber drums before filling

Categories	Defect	Method of Inspection
Critical:	None defined	
Major:	Class AQL 1.00 percent	
101	Foreign material	Visual
102	Gasket missing or damaged	Visual
103	Holes in cover or end	Visual
104	Locking device damaged	Visual
105	Bare area on exterior coating of the chime. The sum of which is in excess of 1 1/2 square inch	Visual
Minor:	AQL 0.65 percent	
201	Poor workmanship, such as: nicks, dents, body bulged or scratches	Visual

4.4.1.3 Applicable to reusable fiber drums before filling

Critical:	None defined	
Major:	100% Inspection	
101	Top chime bent, deformed or cut	Visual
102	Bottom chimes collapsed (annular groove closed or partially closed) or deformed	Visual
103	Body bulged, cut or dented	Visual
104	Gasket in cover missing or damaged	Visual
105	Cover bent, creased or deformed in gasket area or around edge	Visual
106	Locking ring damaged so as to prevent closing	Visual
107	Foreign material, propellant or corrosion	Visual

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(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify "49 CFR 100-199 (latest revision). (ASTM from American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. DEFINITIONS

3.1 Not applicable

4. GENERAL REQUIREMENTS

4.1 Constituent material. -The consistuent materials shall comply with the requirements of the applicable specification as follows:

Constituent Material	Conforming to Specification
Wax, Candelilla	JAN-W-181
Barium Nitrate	MIL-B-162, Class 3
Dibutylphthalate	MIL-D-218
Diethylphthalate	JAN-D-242
Dinitrotoluene	MIL-D-204
Diphenylamine	MIL-D-98
Ethyl Centralite	MIL-E-255, Class 2 or Class 3 (see note)
Graphite	MIL-G-155, Grade III or IV
Nitrocellulose	MIL-N-244, (See Table III)
Nitroglycerin	MIL-N-246, Type I
Nitroguanidine	MIL-N-494, Class to be specified in contract
Potassium Nitrate	MIL-P-156, Class 2 or 3
Potassium Sulfate	MIL-P-193, Type I
2-Nitrodiphenylamine	MIL-N-3399
Lead Carbonate	MIL-L-18618
Cryolite (Technical) sodium aluminum fluoride.)	COMMERCIAL GRADE

NOTE: Any class permitted when added in solution (1)

4.2 Form and dimension

4.2.1 Flake propellant shall conform to the requirements listed in the detail propellant specification or drawings (see 4.2.5.2).

4.2.2 Sheet propellant shall conform to the requirements listed in the detail propellant specification or drawing.

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4.2.3 Grain

4.2.3.1 Type I. -The grain shall be cylindrical with 7 longitudinal perforations, one in the center of the grain and six at the vertices of a symmetrical hexagon.

4.2.3.2 Type II. -The grain shall be cylindrical with a single longitudinal perforation through the center of the grain.

4.2.3.3 The following requirements regarding grain dimensions shall govern unless authorization for departure is given by the procuring activity concerned prior to manufacture of the propellant.

4.2.3.3.1 Length: Diameter Ratio.

4.2.3.3.1.1 Type I. -The average grain length (L) shall be from 2.10 to 2.50 times the average grain diameter (D).

4.2.3.3.1.2 Type II. -The average grain length (L) shall be from 3.0 to 6.0 times the average grain diameter (D).

4.2.3.3.1.3 The length and diameter of grain shall comply with either the mean variation or the standard deviation uniformity requirements shown in Table I.

TABLE I

Mean variation and standard deviation of individual dimensions expressed as a percentage of the mean dimension.

Acceptance Criterion Dimensions	Percent Mean Variation Maximum	Standard Deviation Maximum
Length	6.25	6.25
Diameter (grains 0.2 inch or more in diameter)	3.125	4.75
Diameter (grains less than 0.2 inch in diameter)	6.25	6.25

4.2.3.3.2 Grain diameter. -perforation diameter ratio.

4.2.3.3.2.1 Type I. -The average grain diameter (D) shall be from 5.0 to 15 times the average diameter of the perforation (d).

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